

TIC STEP-BY-STEP SETUP

PREPARATION

1. Retrieve the bankers box labelled Turkish Ice Cream Robot, September 2025 from the MItHRIL shelf. The box includes spare parts as well as the parts needed to setup the demo.
2. Retrieve the Manfrotto tripod from the MItHRIL shelf. Keep the tripod legs at an unextended length.

HARDWARE

1. Using a hex driver, screw in the mounting piece on to the end-effector using the 4 screws pictured below. You will find these in the bankers box.



The end-effector should now look like this:



2. Insert the black carbon fibre rod into the white 3d printed part. Firmly hold onto the rod with one hand, and with the other hand press on the white part until it cannot go in further. **DO NOT ATTEMPT TO INSERT ROD WHILE HOLDING THE MOTOR, HOLD FROM ROD.**

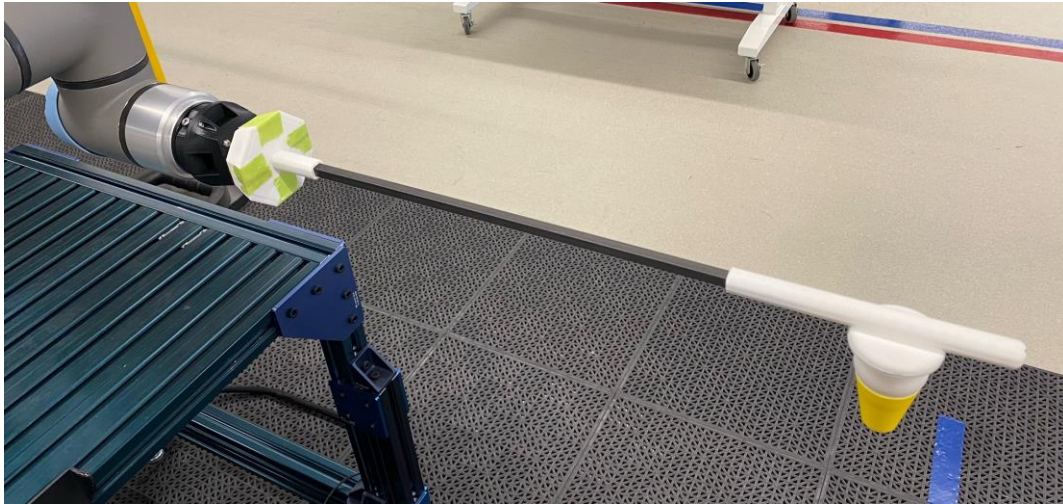


The two parts should now look like this:



3. Now retrieve the cone shaped 3D printed part from the bankers box. It should have a yellow base and a white top with 3 magnets. Align the magnets from the cone and the rod, it should hold.

4. Now align the 3 motor mount screws with the mount on the end-effector. Using a hex driver, securely tighten the motor mount/rod to the end-effector mount. The end-effector should now look like this:





*Replace green tape if broken.

5. Retrieve a Leap 2 camera from the MItHRIL shelf. There will be a black 3D printed part that will fit around the camera in the bankers box. Look for the tripod head in the bankers box as well. Pictured are the three parts in order of occurrence from left to right:

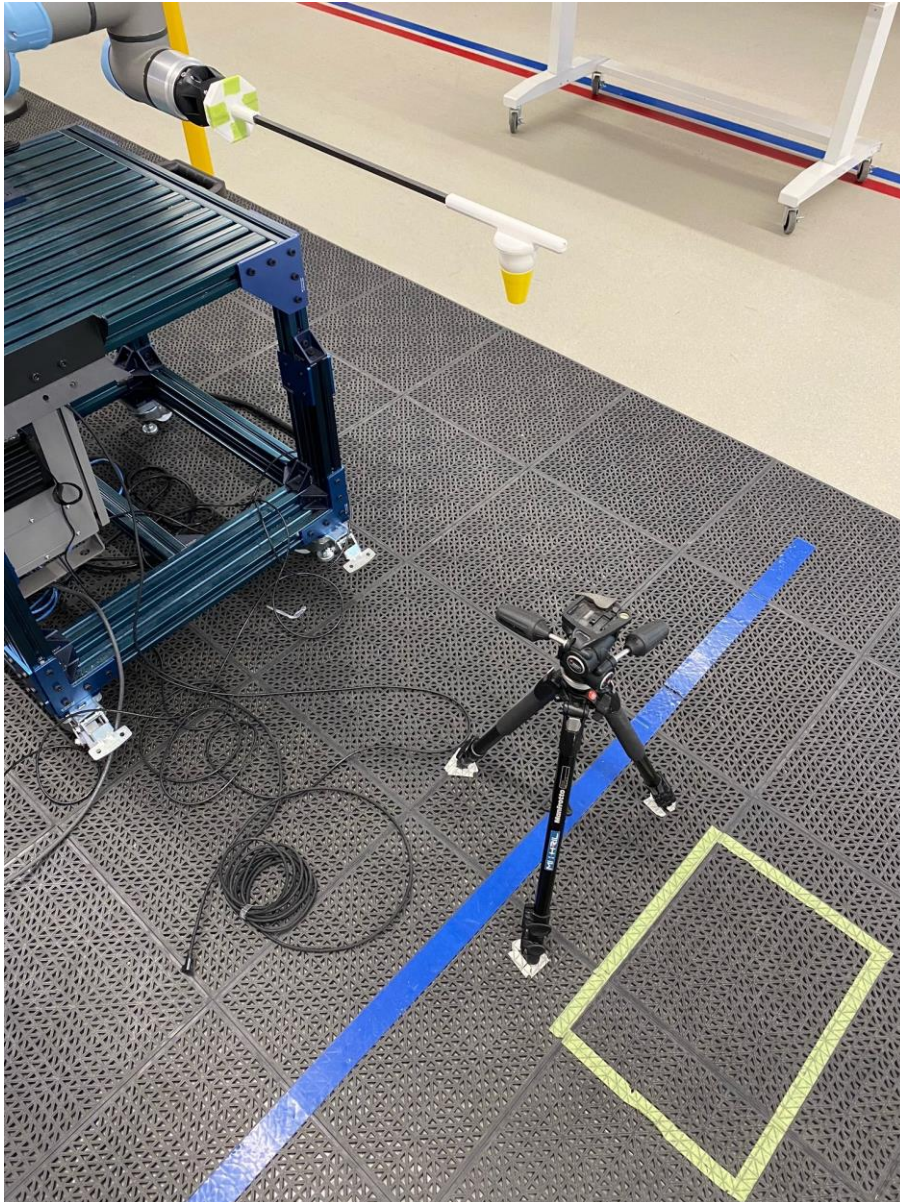


6. Screw in the tripod head to the bottom (flat side) of the black 3D printed part. Carefully push the camera into the black part. PLACE CAMERA LENS SIDE OUT, USB-C PORT SHOULD BE ACCESSIBLE. DO NOT FORCE CAMERA IN PLACE, THERE WILL BE A GAP BETWEEN THE CAMERA AND THE PART. See below:



7. Align the bottom of the tripod to the tape markers on the floor as shown below:





If for any reason the robot has moved, or the markers have been removed. Set the camera to be these distances away from the robot base, the axis direction can be found on the robot tablet:

$x=0.1298$ m

$y=-1.4523$ m

$z=-0.1088$ m

8. Mount the camera on the tripod. It should look like this:

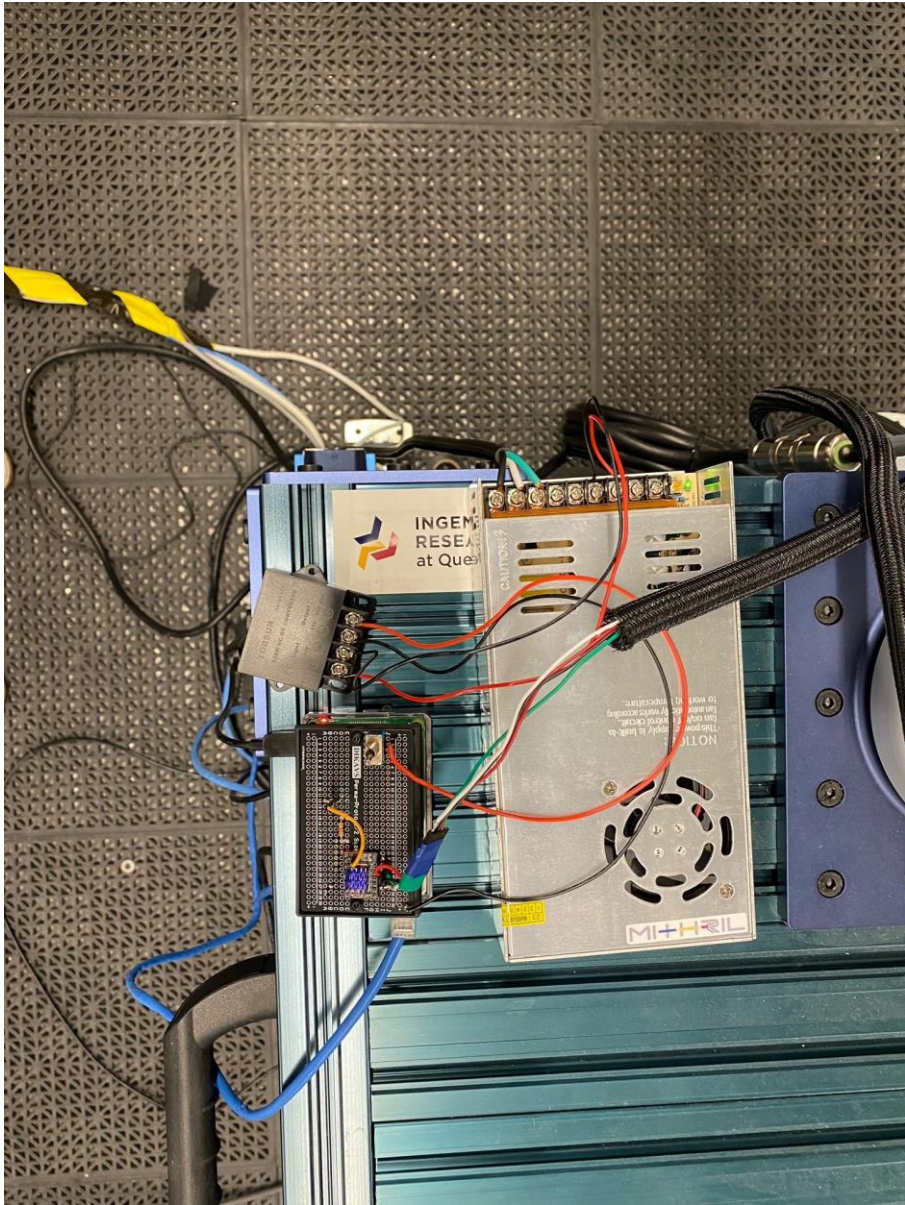


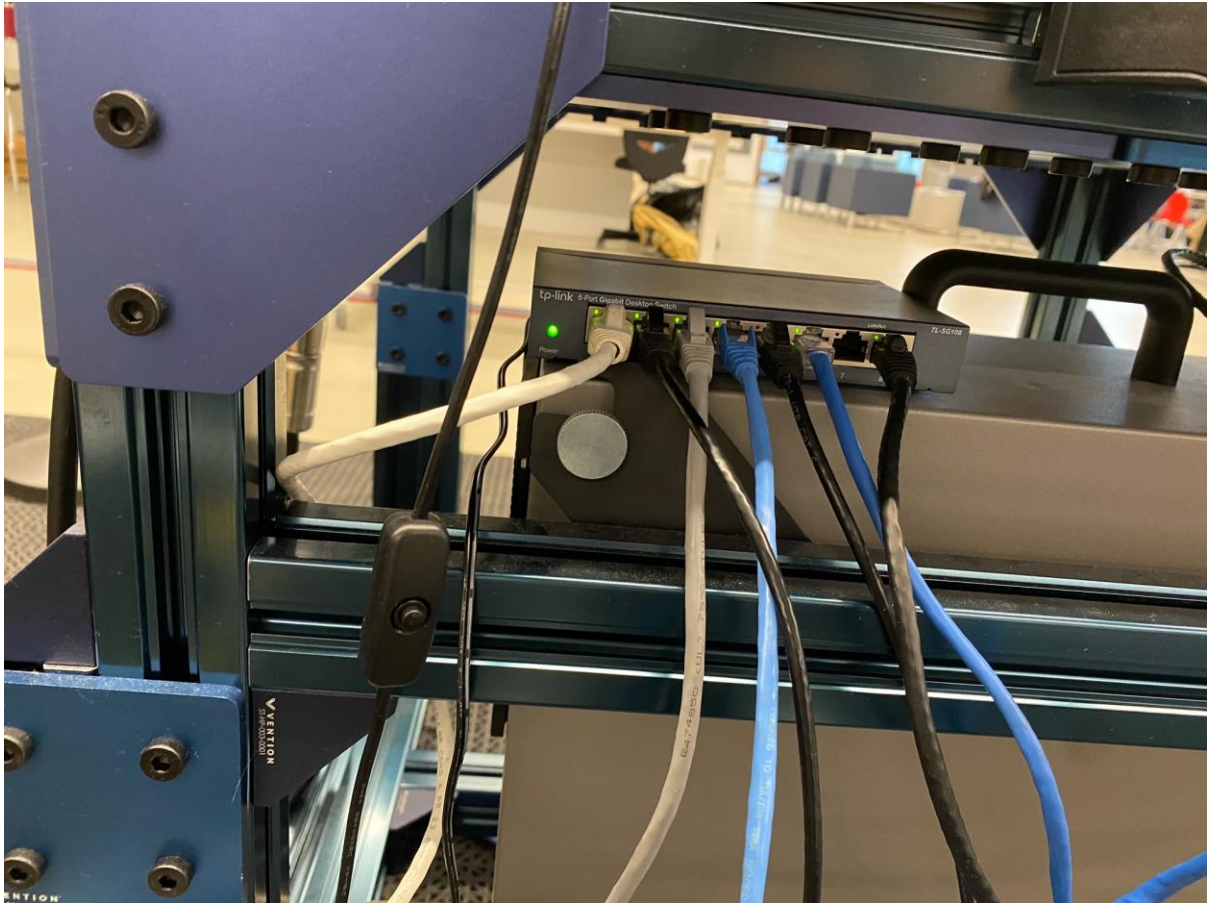
9. Plug in the usb-c cable coming from the Intel NUC under the robot base into the camera. There should be a green light next to the port, and two red lights on the lens.

The setup should now look like this:



10. Near the base of the robot there should be a raspberry pi board, a step-down voltage converter, and a power source block. This is how they are connected:





11. Make sure the power source, the raspberry pi, the port switch under the base, the robot itself, and the intel NUC are all connected to power. If the step-down voltage converter is not connected to the power source block, connect the black (-) and red (+) wires to the corresponding cable clamps on the power source block, do this before connecting the power. View the above pictures.

12. Plug the motor wire into the motor mount port. It is recommended that you use zip ties to tie the motor wire onto the robot with proper slack to ensure it will not tug and break the wires.

HARDWARE SETUP COMPLETE

STEPS TO RUN DEMO

1. Turn on the robot tablet, wait for it to load. To start the robot, click the red circle on the bottom left of the screen and click on, wait for a little bit, then click start. If there are 5 green check marks, the robot has successfully turned on.

2. Make sure that the top right of the screen is set to remote. If it says local, click and switch to remote.

3. Press and hold the intel NUC power button to restart it. It is powered on when the lights are blue.

4. Re-start or start the raspberry pi, click the button along the power cord. Flickering lights indicate the pi is turned on.
5. On the designated computer, login using the MItHRIL password. Click CTRL+ALT+T to open a fresh terminal. On the terminal right click on the black space, hover over Layouts... and click tic_terminal.
6. This is how it should look like after a correct setup:

```
mithril@mithril-Precision-3680: ~/turkish-icecream/test_policies
/bin/bash 38x11
Now in UR5e servo control loop!

None 39x11
Leap Motion Republisher Initialized

root@mithril-pi4: ~ 38x11
than a week old.
To check for new updates run: sudo apt
update
New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to
it.

Last login: Wed Jan  7 13:49:20 2026 f
rom 192.168.1.32
root@mithril-pi4:~#

mithril@mithril-Precision-3680: ~/turkish-icecream
mithril@mithril-Precision-3680:~/turkis
h-icecream/statemachine_dev$

None 39x5
Commands:
b - Move backward
q - Quit
```

7. In the bottom left terminal, hit the up arrow, or alternatively type in “turn_on_motor” and hit enter. The terminal should now look like this:

```
mithril@mithril-Precision-3680: ~/turkish-icecream/test_policies
/bin/bash 38x11
Now in UR5e servo control loop!

None 39x11
Leap Motion Republisher Initialized

root@mithril-pi4: ~ 38x11
r - INFO - GPIO Init finished
2026-01-07 16:21:00,348 - StepperDrive
r - INFO - Init finished
2026-01-07 16:21:00,349 - TMC2209 0 -
INFO - clearing GSTAT
2026-01-07 16:21:00,382 - TMC2209 0 -
INFO - TMC220x Init finished
2026-01-07 16:21:00,383 - TMC2209 0 -
INFO - TMC2209 Init finished
True

mithril@mithril-Precision-3680: ~/turkish-icecream
mithril@mithril-Precision-3680:~/turkis
h-icecream/statemachine_dev$

None 39x5
Commands:
b - Move backward
q - Quit
```


8. In the middle right terminal type the following command: “python3 run_cond_3.py”. This will start the main code.

9. The terminal will prompt you. At this point you should set the cone’s initial position, yellow part facing down, white part facing up. Input “y” and enter in the terminal and the robot will move the cone closer to the designated computer for you to adjust the cone position.

10. Once ready, click enter again. The robot will move away and position itself in the starting position, click enter once more. The demo should now be ready.

11. Hit ctrl+c in the terminal at any time to stop.

TROUBLESHOOT

Use these tips in no particular order to diagnose issues.

1. To do while attempting to run demo. In a new terminal type this command and hit enter: “ros2 topic echo hand_pose”. After waving your hand above the camera, it should return something like this:

header:

stamp:

sec: 1767821731

nanosec: 902123495

frame_id: right

point:

x: -0.1521118682533302

y: -1.5271414428889445

z: -0.025848066486004788

If not, there is an issue with the camera. Try unplugging the usb on the intel NUC side and plugging it back in. Restart the main terminal and try again.

2. If you get a timeout error on the top left terminal, close all terminals and try again, do this a couple of times. If the issue persists, restart the robot, ensure it is turned on and set to remote.
3. If you get a connection error on the bottom left terminal, restart the raspberry pi and try again.
4. If the motor spins are not smooth and seem to be jittery, there may be an issue with the motor wire connection. Check the connection on both ends of the wires, ensure all wires are pushed in correctly and try again.
5. When in doubt check all connections, restart everything and try again.

